How Pruning Criteria Shape Post-RLVR Recovery in Reasoning Language Models

Anonymous ACL submission

Abstract

Structured pruning of reasoning Large Language Models (LLMs) is typically followed by Reinforcement Learning with Verifiable Rewards (RLVR) to recover reasoning ability. The standard pipeline treats the pruning decision as independent of the downstream RL alignment stage. We demonstrate that this independence assumption breaks down: recovery capacity is shaped at pruning time, not during subsequent fine-tuning. Through a controlled comparison of four structural pruning criteria under identical Group Relative Policy Optimization (GRPO) training on DeepSeek-R1-Distill-Qwen-7B, we observe a recovery capacity spectrum: CKA-only pruning achieves active learning (peak reward 0.354), Random stagnates (0.190), Standard exhibits borderline collapse (0.081), and RASP collapses permanently (0.061). We trace the mechanism to preserved representational diversity: CKA-only produces the lowest dense fidelity (μ=0.62 diagonal CKA) yet recovers best, while importance-based methods cluster near μ=0.90 and collapse (Welch's t-test, $p<0.001$). Reward-accuracy decoupling is sparsity-dependent: at 15% sparsity, GRPO amplifies the CKA-only advantage from 5.69 pp (SFT) to 7.20 pp (3-seed GRPO; $p<0.001$), while at 20–25% GRPO cannot improve upon SFT baselines ($|\Delta| < 1$ pp across all methods). This motivates reward trajectory, not accuracy, as the appropriate diagnostic for pruned model quality. Cross-scale (1.5B) and cross-architecture (Qwen2.5-Math-7B) experiments provide initial evidence for generality. Our results demonstrate that pruning criterion selection is an important factor in post-pruning RL trainability, and that CKA-based redundancy scoring consistently outperforms gradient importance for RL-intended compression.

1 Introduction

The choice of which structures to prune from a reasoning LLM shapes whether the model sustains reinforcement learning alignment. Current compression pipelines treat this choice as independent of downstream training: structured pruning removes entire attention heads or MLP channels to reduce memory and latency (Ma et al., 2023; Wang et al., 2026), Supervised Fine-Tuning (SFT) recovers task accuracy, and Reinforcement Learning with Verifiable Rewards (RLVR) restores reasoning quality (DeepSeek-AI, 2025; Shao et al., 2024). Each stage is optimized in isolation. We demonstrate that this separation fails: the pruning criterion significantly influences what we term **recovery capacity**, the model's ability to sustain RL-based recovery without collapsing into degenerate outputs.

The consequences of ignoring this coupling are severe: GRPO training on pruned models can trigger complete collapse, with gradient norms approaching zero, repetitive outputs, and monotonically degrading reward. Whether collapse occurs depends entirely on which structures were removed. Reward itself is an unreliable diagnostic: on unpruned DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), 100 steps of Group Relative Policy Optimization (GRPO; Shao et al., 2024) raise mean reward from 0 to 0.669 while greedy accuracy on GSM8K (Cobbe et al., 2021) remains exactly 81.27%.[1] This reward-accuracy decoupling motivates our focus on recovery capacity as the appropriate diagnostic for post-pruning RL trainability. Ji et al. (2026) claim that RLVR recovers pruned reasoning models to near-original performance, but their evaluation uses a single pruning method and reports only final-checkpoint accuracy without examining training dynamics.

Existing structured pruning methods for reasoning LLMs (Wang et al., 2026, 2025; Ma et al., 2023;

[1]The unpruned model achieves 90.83% without fine-tuning. Applying the same SFT recovery protocol designed for pruned models reduces accuracy to 81.27%, consistent with catastrophic forgetting when further fine-tuning an already well-optimized model.

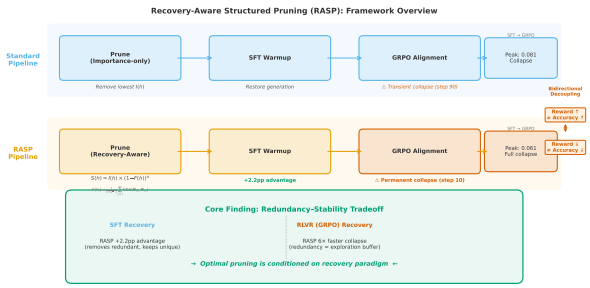


Figure 1: Controlled comparison framework. All four pruning criteria share the same structural analysis (Taylor importance + CKA overlap) and identical recovery pipeline (SFT warmup → GRPO). Despite identical training, the criteria produce divergent outcomes: CKA-only achieves active learning while Standard and RASP collapse.

Kolawole et al., 2025; He et al., 2025; Liu et al., 2025b) optimize for post-pruning quality without studying downstream alignment; none compare multiple pruning criteria under RL or study how pruning affects RL training dynamics. Matsutani et al. (2026) show that RL and SFT impose fundamentally different demands on model representations (RL compresses solution paths while SFT expands them), suggesting that pruning decisions should substantially affect RL trainability, yet no prior work has tested this hypothesis.

We present a controlled comparison of four structural pruning criteria under identical GRPO training on DeepSeek-R1-Distill-Qwen-7B at 20% structured sparsity, with cross-architecture validation on Qwen2.5-Math-7B (Figure 1). The four criteria span a spectrum of representational diversity preservation: **CKA-only** scoring removes structures with the highest pairwise Centered Kernel Alignment (CKA; Kornblith et al., 2019) overlap, preserving all representationally unique structures; **Random** scoring provides an unbiased baseline; **Standard** importance-only scoring retains structures with the highest gradient-based importance (Ma et al., 2023); and **Recovery-Aware Structured Pruning** (RASP) combines importance with CKA overlap in a composite score. Under identical GRPO configuration, these four methods reveal a **recovery capacity spectrum**. CKA-only pruning achieves *active learning*: peak reward reaches 0.354 (4.4× Standard). Random pruning stagnates (reward oscillates 0.14–0.19). Standard pruning exhibits borderline collapse (peak 0.081). RASP collapses permanently (peak 0.061, 85% of completions yield zero reward variance). The ordering, CKA-only ≫ Random > Standard > RASP, follows the degree of preserved representational diversity. Gradient importance scoring, the standard criterion for LLM pruning, removes structures scored "unimportant" by gradient metrics that are in fact representationally unique, reducing the exploration diversity available for RL training in our setting. A depth pruning baseline (ShortGPT; Men et al., 2024) is the only method where GRPO actively degrades accuracy (-1.36 pp), confirming that pruning granularity interacts with RL trainability.

Our contributions are:

1. We discover a **recovery capacity spectrum** across four pruning criteria under controlled GRPO training: CKA-only enables active learning (peak reward 4.4× Standard), Random stagnates, and Standard/RASP collapse. Extended training (300 steps) confirms this gap is structural, not a matter of training budget (§4.2, §4.5).
2. We show that **reward-accuracy decoupling** is sparsity-dependent: at 15% sparsity, GRPO amplifies the CKA-only advantage (+1.51 pp differential), while at 20–25% GRPO cannot improve upon SFT baselines. This motivates reward trajectory as the diagnostic for recovery capacity (§4.3).
3. We identify **preserved representational diversity** as a plausible mechanism: CKA-only (μ=0.62 diagonal CKA) diverges most from the dense model, while collapsing methods preserve dense fidelity ($\mu \geq 0.88$). Cross-architecture (Qwen2.5-Math-7B) and cross-scale (1.5B) experiments provide initial evidence for generality (§4.5).

2 Related Work

Structured pruning of LLMs. LLM-Pruner (Ma et al., 2023) introduced dependency-aware structural pruning for transformers; Bonsai (Kolawole et al., 2025) achieves competitive results using only forward passes; GRASPrune (Wang et al., 2026) advances the state of the art with global gating that jointly prunes FFN channels and KV head groups. ShortGPT (Men et al., 2024) and SliceGPT (Ashkboos et al., 2024) target depth pruning, while Wanda (Sun et al., 2024) addresses unstructured sparsity; Monjur et al. (2026) revisit the structured-vs-unstructured trade-off under test-time scaling. For reasoning models, RESP (Wang et al., 2025)

uses gradient-based importance on self-generated reasoning traces, RAC (Lucas et al., 2025) reconstructs chain-of-thought during pruning, SSGR (Xiang et al., 2025) uses self-generated calibration data, and Ding et al. (2026) find width pruning more robust than depth pruning for reasoning but do not study RLVR recovery. A parallel line recognizes that functional structure should guide pruning: FANG (Yu et al., 2025) groups FFN neurons by activation clusters, FuncNet (Liu et al., 2025a) decomposes LLMs via ICA, and MPruner (Hu et al., 2024) uses inter-layer CKA (Kornblith et al., 2019) for depth pruning in CNNs. Our work uses *intra-layer* CKA between individual structures at a finer granularity, optimizing for RL recovery capacity rather than immediate accuracy preservation.

Pruning-aware recovery. Ji et al. (2026) demonstrate that RLVR restores pruned reasoning models to near-original accuracy, significantly outperforming SFT, but test only one pruning method without studying whether the pruning criterion affects recovery. P^2 Law (Chen et al., 2025) establishes scaling laws for post-training loss after pruning. PASER (He et al., 2025) optimizes recovery data selection, PAT (Liu et al., 2025b) jointly considers pruning and fine-tuning objectives, RestoreLCC (Feng et al., 2025) contrastively probes critical attention heads for post-hoc compensation, Zhao et al. (2026) adopt iterative prune-tune loops for gradual compression, and OPSD (Kim and Lee, 2026) compresses reasoning models *after* RLVR (the reverse of our pipeline). These approaches optimize complementary axes (data, training, compensation, schedule); our work isolates which *pruning criterion* enables RL-based recovery. The Lottery Ticket Hypothesis (Frankle and Carbin, 2019) asks which subnetworks train effectively from random initialization; our recovery capacity spectrum extends this to *paradigm-conditioned trainability*, where the optimal subnetwork depends on the recovery paradigm (RLVR vs. SFT).

RLVR training dynamics. Matsutani et al. (2026) show that RLVR *compresses* incorrect reasoning trajectories via global exploration, while SFT *expands* correct ones via local imitation. This asymmetry motivates our finding: gradient importance scoring protects structures that RL would compress, while removing the diverse representations RL exploration requires. RL training instability has been studied in the RLHF context: Gao et al. (2023) characterize reward overoptimization, Dong et al. (2025) document repetitive output collapse during extended RL training, and Shah et al. (2025) analyze compounding bias from reward estimation. Our work connects this instability literature to structural pruning, showing that the pruning criterion shapes susceptibility to RL collapse.

3 Method

We design a controlled comparison framework to isolate the effect of pruning criterion on post-RLVR recovery capacity. The framework consists of three components: (1) shared structural analysis that computes importance and functional overlap scores for all prunable structures, (2) four scoring criteria that combine these scores differently to select structures for removal, and (3) an identical two-stage recovery pipeline applied to all pruned models. By varying only the scoring criterion while holding all other factors constant, we can attribute differences in GRPO training dynamics directly to the pruning decision.

3.1 Preliminaries and Notation

Structured pruning. Given a pretrained transformer $\mathcal{M}$ with L layers, structured pruning removes entire computational units (attention heads or MLP channels) to reduce model size while maintaining hardware-friendly dense computation. We denote the set of prunable structures as $\mathcal{H} = \{h_1, h_2, \ldots, h_N\}$.

Grouped Query Attention (GQA). In GQA architectures (Ainslie et al., 2023), n_Q query heads share n_{KV} key-value heads, with each KV head serving a group of $G = n_Q/n_{KV}$ query heads. For DeepSeek-R1-Distill-Qwen-7B (n_Q=28, n_{KV}=4), $G = 7$, yielding 4 GQA groups per layer. We treat each GQA group (7 Q heads + 1 KV head) as a single prunable unit, since removing a KV head necessitates removing all associated Q heads.

RLVR and GRPO. Reinforcement Learning with Verifiable Rewards (RLVR) recovers pruned model capabilities using outcome-based rewards from tasks with verifiable answers. We use Group Relative Policy Optimization (GRPO; Shao et al., 2024) as the recovery algorithm.

3.2 Structural Analysis

All four pruning criteria share a common structural analysis stage that computes two scores per struc-

ture: *importance* $I(h)$ capturing the structure's contribution to reasoning, and *functional overlap* $F(h)$ measuring representational redundancy with other structures in the same layer.

Calibration data. We compute importance and overlap scores on chain-of-thought reasoning data from GSM8K (Cobbe et al., 2021) training examples. Given a calibration set $\mathcal{D}_{\text{cal}} = \{(x_j, y_j)\}_{j=1}^{M}$, we compute activations and gradients on the concatenated sequences (x_j, y_j).

Structural importance. For each structure h, we compute importance $I(h)$ via first-order Taylor expansion on the reasoning trace loss:

$$I(h) = \left| \sum_{(x,y)\in\mathcal{D}_{\text{cal}}} \mathbf{a}_h(x,y)^\top \frac{\partial \mathcal{L}(x,y)}{\partial \mathbf{a}_h} \right|, \quad (1)$$

where $\mathbf{a}_h$ denotes the activation of structure h and $\mathcal{L}$ is the language modeling loss over the reasoning trace. For a GQA group g containing Q heads $\{q_1, \ldots, q_G\}$ and KV head k, we aggregate: $I(g) = \sum_{i=1}^{G} I(q_i) + I(k)$.

Functional overlap via CKA. We measure functional overlap using linear Centered Kernel Alignment (CKA; Kornblith et al., 2019), which quantifies similarity between two sets of representations invariant to invertible linear transformations. Given activation matrices $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n\times d}$ collected over n calibration examples, linear CKA is:

$$\text{CKA}(\mathbf{X}, \mathbf{Y}) = \frac{\text{HSIC}(\mathbf{X}, \mathbf{Y})}{\sqrt{\text{HSIC}(\mathbf{X}, \mathbf{X}) \cdot \text{HSIC}(\mathbf{Y}, \mathbf{Y})}}, \quad (2)$$

where $\text{HSIC}(\mathbf{X}, \mathbf{Y}) = \frac{1}{(n-1)^2} \operatorname{tr}(\mathbf{KHLH})$, with $\mathbf{K} = \mathbf{X}\mathbf{X}^\top$, $\mathbf{L} = \mathbf{Y}\mathbf{Y}^\top$, and $\mathbf{H} = \mathbf{I}_n - \frac{1}{n}\mathbf{1}\mathbf{1}^\top$.

For a GQA group g_i in layer ℓ, we form its representation by concatenating its constituent Q head activations: $\mathbf{X}_{g_i} = [\mathbf{a}_{q_1}; \ldots; \mathbf{a}_{q_G}] \in \mathbb{R}^{n\times G d_h}$. The functional overlap of g_i is the mean pairwise CKA with other groups in the same layer:

$$F(g_i) = \frac{1}{|\mathcal{G}_\ell| - 1} \sum_{\substack{g_j \in \mathcal{G}_\ell \\ j \neq i}} \text{CKA}(\mathbf{X}_{g_i}, \mathbf{X}_{g_j}). \quad (3)$$

We apply the same intra-layer CKA overlap to MLP channel groups, partitioning each MLP intermediate dimension into equal-sized groups. Unlike prior work that uses inter-layer CKA to identify globally redundant layers for depth pruning (Hu et al., 2024; Lachance, 2022), we use *intra-layer* CKA between individual structures to measure per-structure functional redundancy. Computing all pairwise CKA scores adds negligible overhead: with at most 28 structures per layer and 28 layers, the total number of CKA evaluations is $O(L \cdot k^2)$ where $k \leq 28$, completing in under 2 minutes on a single GPU.

CKA reliability. Our intra-layer use of linear CKA between structures of identical dimensionality avoids the kernel width and layer width sensitivities identified by Davari et al. (2023); Klabunde et al. (2023); empirically, structure rankings are stable across calibration sizes (Spearman $\rho \geq 0.964$; Appendix H).

3.3 Pruning Scoring Criteria

We compare four scoring criteria that combine importance $I(h)$ and functional overlap $F(h)$ in different ways. All four methods use the same per-layer capacity floor ($\geq$40% of MLP groups per layer must be retained, chosen to prevent any single layer from becoming a throughput bottleneck) and global ranking procedure: structures are sorted by score, and the lowest-scoring structures are pruned up to the target sparsity ratio, skipping any candidate whose removal would violate the layer floor.

Standard (importance-only). Following LLM-Pruner (Ma et al., 2023), structures are ranked solely by importance:

$$S_{\text{std}}(h) = I(h). \quad (4)$$

This is the industry-standard criterion that retains gradient-important structures and removes "unimportant" ones.

RASP (recovery-aware composite). The recovery-aware score combines importance and functional overlap:

$$S_{\text{rasp}}(h) = I(h) \times \big(1 - F(h)\big)^{\alpha}, \quad (5)$$

where $\alpha \geq 0$ controls the strength of the overlap penalty (α=0 reduces to Standard). Structures with high importance *and* low overlap receive the highest scores and are retained; structures with high overlap are penalized and pruned preferentially, regardless of their importance.

CKA-only (overlap-only). CKA-only scoring ignores gradient importance entirely, ranking structures by functional overlap alone. Structures with

the highest mean pairwise CKA similarity to other structures in their layer are removed first:

$$S_{\text{cka}}(h) = 1 - F(h). \tag{6}$$

This criterion preserves all representationally unique structures regardless of their gradient importance, removing only those whose function is redundant with retained structures. S_{cka} corresponds to the limit $\alpha \to \infty$ in Eq. 5.

Random. Structures are selected uniformly at random for removal, subject to the per-layer capacity floor. This baseline controls for the effect of any informed scoring criterion.

The four criteria span a spectrum from pure gradient importance (Standard) to pure representational diversity preservation (CKA-only), with RASP as a weighted interpolation and Random as an uninformed control. The key distinction is that CKA-only protects *functionally unique* structures regardless of gradient contribution: structures deemed "unimportant" by gradient metrics may provide non-redundant computational pathways essential for RL exploration (Matsutani et al., 2026).

3.4 Two-Stage Recovery

After pruning, all four models undergo an identical two-stage recovery pipeline. Any differences in GRPO training dynamics are therefore attributable to the pruning criterion alone.

Stage 1: SFT warmup. Structured pruning disrupts the model's ability to generate end-of-sequence (EOS) tokens: in sampling mode, pruned models produce repetitive outputs without termination. Since GRPO computes rewards only from completed generations, this EOS collapse makes direct RLVR recovery impossible. We restore generation competence through a short SFT phase on chain-of-thought reasoning demonstrations with teacher forcing. This stage is intentionally brief: its purpose is to restore the minimal capabilities (EOS generation, format compliance) needed for GRPO, not to recover full reasoning ability.

Stage 2: GRPO alignment. With generation competence restored, we apply GRPO on mathematical reasoning problems with binary reward (r=1 if the final answer matches the ground truth, r=0 otherwise). RLVR is preferred over extended SFT for reasoning recovery because of their fundamentally different training dynamics: Matsutani et al. (2026) show that RLVR *compresses* incorrect reasoning trajectories while SFT *expands* correct ones. For a pruned model, RLVR can globally explore and redistribute functionality across remaining structures, while SFT is limited to imitating fixed trajectories through a reduced architecture. The GRPO configuration (learning rate, KL coefficient, sampling temperature, loss formulation) is held constant across all four pruned models (Section 4.1).

3.5 Recovery Capacity Diagnostics

We define **recovery capacity** as the model's ability to benefit from post-pruning recovery training, operationalized in our GRPO experiments as peak mean reward over the training horizon (Table 1). We track four diagnostic measures—mean reward, zero-variance batch fraction (f_0), gradient norm, and KL divergence—to distinguish *active learning* (rising reward, low f_0), *stagnation* (flat reward), and *collapse* (declining reward, high f_0, vanishing gradients).

The complete pruning comparison procedure is given in Algorithm 1 (Appendix B).

4 Experiments

4.1 Experimental Setup

Base model and evaluation. We evaluate on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025), a 7B-parameter reasoning model built on the Qwen2.5 architecture (Yang et al., 2024) with GQA (Ainslie et al., 2023) (architecture details in Section 3.1). We evaluate on GSM8K (Cobbe et al., 2021), a benchmark of 1,319 grade-school math problems requiring multi-step arithmetic reasoning. We report exact-match accuracy: the final numerical answer is extracted from the model's generation and compared against the ground truth. For rapid iteration during development, we use a 100-problem subset (N=100); final results use the full test set (N=1,319). The original unpruned model achieves 90.83% on GSM8K (greedy decoding).

Pruning methods. We compare four pruning criteria at 20% structured sparsity, all removing the same number of MLP channel groups (200 of 784 across 28 layers, ∼1.45B parameters) with identical per-layer capacity floors (≥40% retention per layer):

1. **CKA-only**: structures ranked by functional overlap $F(h)$ alone (Eq. 3); the most redundant groups are removed regardless of gradient importance.

2. **Random**: uniform random selection among eligible groups.
3. **Standard**: Taylor-expansion importance $I(h)$ alone (Eq. 1); the least important groups are removed, following LLM-Pruner (Ma et al., 2023).
4. **RASP** (α=1.0): the composite score $S(h) = I(h) \times (1-F(h))^{\alpha}$ (Eq. 5), removing structures that are both low-importance and functionally redundant.

Recovery pipeline and hardware. All four methods undergo matched two-stage recovery. Stage 1 (SFT warmup): 2 epochs on 7,473 GSM8K chain-of-thought examples with teacher forcing (37 minutes), restoring EOS generation and basic format compliance. Stage 2 (GRPO): Group Relative Policy Optimization (Shao et al., 2024) with learning rate 1×10^{-6}, maximum completion length 1,024 tokens, KL penalty coefficient β=0.01, sampling temperature 0.4, DAPO-style clipped loss, and binary reward (r=1 if correct, r=0 otherwise), trained for 100 steps on 1,000 GSM8K training problems. All methods share identical GRPO hyperparameters, including `reward_desert_patience`=50. Seeds vary at the GRPO stage; pruning calibration and SFT use identical configurations across all methods. We repeat GRPO training with 3 random seeds for CKA-only and Standard (Table 2). CKA-only achieves 64.39% $\pm$ 0.19 pp (std across seeds), Standard 59.84% $\pm$ 0.38 pp; the 4.55 pp gap is robust across seeds (p=0.0009), confirming that recovery capacity differences are structural rather than stochastic. All experiments run on 2$\times$ NVIDIA RTX PRO 6000 GPUs (94 GB VRAM each); pruning variants are trained in parallel on separate GPUs under identical wall-clock conditions.

4.2 Recovery Capacity Spectrum

The central finding of this work is that the pruning criterion can significantly influence post-RLVR recovery capacity in our experimental setting. Under controlled GRPO training with identical hyperparameters, the four pruning methods produce qualitatively distinct training dynamics that form a clear recovery capacity spectrum: CKA-only $\gg$ Random $\gg$ Standard $>$ RASP. Table 1 summarizes the key metrics, and Figure 2 shows the reward trajectories over 100 GRPO steps.

CKA-only pruning is the only method that produces active learning under GRPO. Reward reaches a peak of 0.354 (mean 0.285) with f_0 as low as 0.125, including a recovery event at step 90 where reward rebounds from a transient perturbation. Random pruning produces stagnation: reward oscillates between 0.142 and 0.190 with f_0 near 0.33.

Standard pruning exhibits borderline collapse (Figure 3): reward peaks at 0.081 (step 40) then oscillates near 0.05, with $f_0 \geq 0.500$ throughout—more than half of batches produce no learning signal. RASP produces full collapse: reward peaks at 0.061 (step 50) then declines to 0.010 by step 100, with $f_0 > 0.675$ and clipped ratio $> 90\%$.

The 5.8$\times$ peak reward gap (CKA-only 0.354 vs RASP 0.061) arises from models pruned to identical parameter budgets with identical GRPO configurations; the only variable is which structures were removed.

Table 1: Recovery capacity gradient: GRPO dynamics for four pruning criteria at 20% sparsity under identical configuration (lr 1×10^{-6}, β=0.01, T=0.4, 100 steps). Peak reward = best across 100 training steps. f_0: fraction of training batches where all sampled completions receive identical reward (lower = healthier exploration). Best in **bold**, second-best underlined.

Method	Peak Rew. ↑	Mean Rew. ↑	Min f_0 ↓	Behavior
CKA-only	**0.354**	**0.285**	**0.125**	Active learning
Random‡	<u>0.190</u>	<u>0.164</u>	<u>0.300</u>	Stagnation
Standard	0.081	0.050	0.500	Borderline collapse
RASP‡	0.061	0.031	0.675	Full collapse

‡Single seed (N=1). CKA-only and Standard validated with 3 seeds (Table 2).

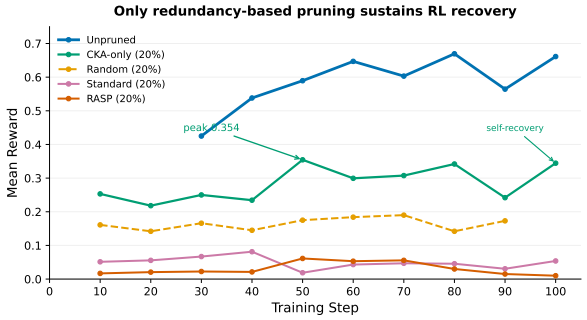


Figure 2: GRPO reward trajectories for four pruning criteria and the unpruned model under identical training configuration (β=0.01, T=0.4, 100 steps). CKA-only reaches 0.354 peak reward with an observed recovery event at step 90. Random oscillates between 0.14 and 0.19 (training timed out at step 97; 9 steps shown). Standard peaks at 0.081 before borderline collapse. RASP peaks at 0.061 (step 50) and declines to 0.010 by step 100.

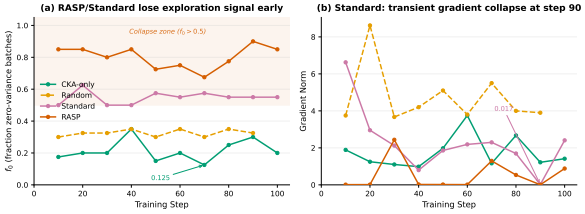


Figure 3: Training dynamics diagnostics. (a) f_0 (fraction of zero-variance batches): CKA-only stays below 0.35; Standard and RASP remain in the collapse zone ($f_0 > 0.5$) throughout training. (b) Gradient norm: Standard exhibits a transient minimum of 0.017 at step 90 before recovering to 2.41 at step 100; RASP gradient norms remain predominantly near zero.

Table 2: GRPO responsiveness on GSM8K (N=1,319, greedy decoding). All pruned models use the controlled v5 GRPO configuration (identical hyperparameters including `reward_desert_patience`=50), evaluated at step 100. CKA-only and Standard report mean ± std over 3 GRPO seeds; RASP uses a single seed. Best in **bold**, second-best underlined.

Model	SFT Acc. (%)	GRPO Acc. (%)	Δ (pp)	Peak Rew.
Unpruned	**81.27**	**81.27**	0.00	**0.669**
CKA-only 20%	63.91	64.39±0.19	+0.48	0.354
Standard 20%	59.29	59.84±0.38	+0.55	0.081
RASP 20%	61.49	60.88	−0.61	0.061
Random 20%‡	–	–	–	0.[illegible]
ShortGPT 20%†	44.73	43.37	−1.36	0.[illegible]

†Depth pruning baseline (Men et al., 2024); removes entire layers.
‡Accuracy not evaluated; checkpoint not preserved.

4.3 Reward-Accuracy Decoupling

GRPO reward trajectory reflects training capacity—the model's ability to learn from RL signal—not downstream accuracy. At 20% sparsity, accuracy changes are small across all pruned methods ($|\Delta| < 1$ pp: CKA-only +0.48, Standard +0.55, RASP −0.61 pp; Table 2) despite the 4.4× peak reward gap.

The degree of decoupling varies with sparsity. At 15% (3-seed GRPO), SFT establishes a 5.69 pp gap between CKA-only and Standard (72.18% vs 66.49%; single SFT seed), which GRPO maintains or slightly amplifies to 7.20 pp (72.76% vs 65.56%; Welch's t-test, $p<0.001$). At 20%, the differential GRPO contribution is negligible: SFT establishes a 4.62 pp gap that GRPO preserves (4.55 pp). At 25% (single seed), decoupling is most extreme: Standard's peak reward collapses to 0.008 yet accuracy changes by only +0.99 pp (Appendix L). Across all four fine-grained pruning criteria and both sparsity levels tested (20% and 25%), $|\Delta| < 1$ pp without exception on GSM8K (Tables 2, 12), extending to MATH-500 where all completed SFT/GRPO pairs show $|\Delta| \leq 1.2$ pp (Appendix F). This pattern suggests that moderate structural damage (15%) preserves enough capacity for GRPO to differentially benefit CKA-only pruning, while more aggressive pruning (20–25%) exceeds the threshold where GRPO can improve upon SFT baselines.

Extended training to 300 steps at 20% confirms: accuracy remains unchanged (CKA-only 64.37%→64.37%, Standard 59.44%→59.29%; single seed). The pruning-criterion ranking transfers to MATH-500 (Hendrycks et al., 2021) (CKA-only 26.6% vs Standard 25.8% vs RASP 22.6%; Appendix F), and decoupling holds across HumanEval (Chen et al., 2021) and MBPP (Austin et al., 2021) (all pruned-model GRPO $|\Delta| < 2$ pp; Appendix G). ShortGPT (Men et al., 2024) depth pruning is the exception: GRPO interaction is task-dependent (−1.36 pp on GSM8K, +4.01 pp on MBPP; Tables 2, 8), unlike the uniformly neutral deltas of fine-grained methods ($|\Delta| \leq 1.34$ pp).

4.4 Additional Results

Post-SFT accuracy does not predict GRPO outcome. Under SFT alone, CKA-only (63.91%) > RASP (61.49%) > Standard (59.29%), yet under GRPO the RASP/Standard ordering reverses (peak 0.081 vs 0.061). More strikingly, ShortGPT post-SFT (44.73%) shows GRPO actively harming accuracy (−1.36 pp), while unstructured pruning achieves the highest post-SFT accuracy (87.87%) with GRPO accuracy 87.64% (Δ=−0.23 pp), confirming that pruning criteria are irrelevant when structural damage is minimal (96.5% dense retention). Post-SFT accuracy does not monotonically determine GRPO recovery outcome; the recovery capacity spectrum is driven by structural properties of the pruned architecture, not residual SFT quality.

Additional results. All methods collapse at 30% sparsity, validating 20% as the operating point (Appendix J). Pruning reduces parameters by 19% and memory by ∼28% (Appendix C). The CKA coefficient α shows a non-monotonic pattern in post-prune accuracy across $\{0.0, 0.25, 0.5, 1.0, 2.0, 4.0\}$; notably, α=4.0 exhibits active GRPO learning (peak reward 0.256) with preliminary accuracy 62.93%, contrasting with α=1.0's collapse (peak reward 0.061), consistent with CKA dominance driving recovery capac-

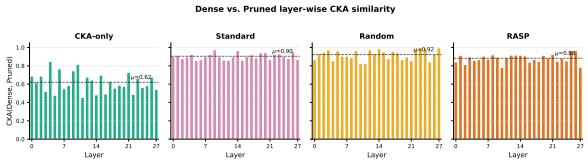


Figure 4: Layer-wise CKA similarity between dense and pruned representations (28 layers). CKA-only diverges most from dense (μ=0.62) yet recovers best; Standard (μ=0.90), Random (μ=0.92), and RASP (μ=0.88) preserve dense fidelity but collapse under RL.

Table 3: Cross-scale validation at 20% sparsity. 7B: DeepSeek-R1-Distill-Qwen-7B (GSM8K mean over 3 seeds for CKA-only and Standard; see Table 2). 1.5B: DeepSeek-R1-Distill-Qwen-1.5B (single seed). GSM8K accuracy (N=1,319). 7B reward = peak across 100 GRPO steps (cf. Table 1); 1.5B reward = mean across training steps.

Method	7B		1.5B	
	Peak Rew.	GSM8K (%)	Rew.	GSM8K (%)
CKA-only	**0.354**	**64.39**	**0.238**	**35.48**
Standard	0.081	59.84	0.084	22.29
RASP	0.061	60.88	0.099	26.16
Random	0.190	–	–	–

ity (Appendix D).[2]

4.5 Extended Analysis

Representational diversity. CKA similarity between dense and pruned representations (Figure 4) shows CKA-only diverges most from the dense model (μ=0.62), while Standard (μ=0.90), Random (μ=0.92), and RASP (μ=0.88) cluster near 1.0. Gradient importance preserves dense fidelity at the cost of redundant representations that provide no exploration benefit during GRPO. Structural overlap analysis reveals that CKA-only concentrates pruning in early layers (L0–9, 72.5% of removed groups) while preserving later layers critical for reasoning (L16–27), whereas Standard prunes uniformly across depth, removing many late-layer structures that CKA identifies as representationally unique (Appendix N).

Extended training cannot rescue collapse. Extending GRPO from 100 to 300 steps (Figure 5, Appendix E), Standard shows zero net improvement (reward 0.054–0.058, $f_0 > 0.5$ throughout), while CKA-only sustains a 5.5× gap (0.308 vs 0.056 at step 300). The gap is structural, not a matter of training budget.

[2] α=4.0 results are preliminary (N=774/1,319 samples evaluated); full evaluation pending.

Cross-scale and cross-architecture generalization. On DeepSeek-R1-Distill-Qwen-1.5B at 20% sparsity (Table 3), the three-method ordering is preserved (CKA-only 35.48% $\gg$ RASP 26.16% $>$ Standard 22.29%), with reward-accuracy decoupling confirmed at this scale ($|\Delta| < 1$ pp: CKA-only −0.15, Standard −0.61 pp). To test cross-architecture generalization, we apply CKA-only and Standard pruning to Qwen2.5-Math-7B (Yang et al., 2024) at 20% sparsity (Table 13, Appendix M). CKA-only causes near-total initial damage (1.0% post-prune accuracy vs Standard's 18.5%), yet SFT recovers CKA-only to 51.50% ± 3.53 while Standard remains at 21.00% ± 2.88 (N=200). This post-prune accuracy reversal—where the method with lower initial accuracy achieves dramatically higher recovery—provides additional evidence that recoverability is shaped by preserved network structure, not starting accuracy. Cross-architecture validation covers the SFT recovery stage; GRPO validation on Qwen2.5-Math-7B is deferred to future work.

5 Discussion and Conclusion

We present a controlled comparison demonstrating that the structural pruning criterion shapes post-RLVR recovery capacity. Under identical GRPO training, CKA-only pruning enables active learning (peak reward 4.4× Standard), while importance-based methods collapse; extended training to 300 steps confirms this gap is structural. The plausible mechanism is preserved representational diversity: CKA-only diverges most from the dense representation (μ=0.62) yet recovers best, while collapsing methods preserve dense fidelity ($\mu \geq 0.88$). Reward-accuracy decoupling is sparsity-dependent: at 15%, GRPO amplifies the CKA-only advantage (5.69 pp→7.20 pp), while at 20–25% GRPO cannot improve upon SFT baselines, motivating reward trajectory as the appropriate diagnostic. Cross-scale (1.5B) and cross-architecture (Qwen2.5-Math-7B) experiments provide initial evidence for generality. Code, pruning masks, and model checkpoints will be released upon acceptance.

Limitations

Architecture and scale. Our primary results are on DeepSeek-R1-Distill-Qwen-7B (DeepSeek-AI, 2025) with GQA (G=7). Cross-architecture validation on Qwen2.5-Math-7B covers SFT recovery only (N=200); both models share the Qwen2.5

base architecture, so our cross-architecture evidence does not extend to different architecture families (e.g., LLaMA, Mistral). Whether the spectrum generalizes across architecture families and scales (1B–70B) remains open.

Diagnostic limitations. Reward-accuracy decoupling at 20–25% sparsity means that recovery capacity—operationalized as peak reward—does not predict downstream accuracy improvements within our training horizon. Recovery capacity captures the model's ability to sustain RL training without collapse, but practitioners seeking accuracy gains may require longer training, different reward formulations, or lower sparsity levels. The per-layer capacity floor (40% retention) was chosen heuristically and not ablated; different floor values may shift the recovery capacity spectrum.

Replication and scope. CKA-only and Standard are validated over 3 GRPO seeds; RASP and Random each use a single seed (N=1). We evaluate on four benchmarks (GSM8K, MATH-500, HumanEval, MBPP) covering math reasoning and code generation; logical, commonsense, and multimodal reasoning are not addressed.

Ethics Statement

This work uses publicly available models (DeepSeek-R1-Distill-Qwen, Qwen2.5-Math) and established benchmarks (GSM8K, MATH-500, HumanEval, MBPP); no private data or human subjects are involved. Structured pruning reduces model size and inference memory, which can lower the computational cost and energy consumption of deploying large language models. Our research focuses on the technical analysis of pruning criteria and their interaction with reinforcement learning recovery; we do not foresee direct negative societal impacts from this work.

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A Training Details

Table 4: Hyperparameter details for SFT warmup and GRPO recovery stages.

Parameter	SFT	GRPO
Learning rate	2×10^{-5}	1×10^{-6}
Batch size	1	1
Gradient accumulation	8	1
Max sequence length	2,048	–
Max new tokens	–	1,024
Optimizer	AdamW	AdamW
Epochs / Steps	2 epochs	100 steps
KL coefficient β	–	0.01
Sampling temperature	–	0.4
Num generations per prompt	–	8
Reward	–	Binary (exact match)
Loss formulation	Cross-entropy	DAPO-style clipped
`reward_desert_patience`	–	50

B Algorithm

Algorithm 1 Controlled Pruning Comparison Framework

Require: Model $\mathcal{M}$, calibration data $\mathcal{D}_{\text{cal}}$, target sparsity s, scoring criterion $\mathcal{C} \in \{\text{Std}, \text{RASP}, \text{CKA}, \text{Rand}\}$, layer floor τ
Ensure: Pruned and recovered model $\mathcal{M}'$

% Stage 1: Shared Structural Analysis
for each structure $h \in \mathcal{H}$ **do**
  Compute importance $I(h)$ via Eq. (1)
  Collect activations $\mathbf{X}_h$ on $\mathcal{D}_{\text{cal}}$
for each layer $\ell = 1, \ldots, L$ **do**
  for each structure h_i in layer ℓ **do**
    $F(h_i) \leftarrow \frac{1}{|\mathcal{H}_\ell|-1} \sum_{j \neq i} \text{CKA}(\mathbf{X}_{h_i}, \mathbf{X}_{h_j})$
% Stage 2: Criterion-Specific Scoring
for each structure $h \in \mathcal{H}$ **do**
  $S(h) \leftarrow \mathcal{C}(I(h), F(h))$ ▷ Eq. 4, 5, 6, or random
Sort $\mathcal{H}$ by $S(h)$ ascending; prune lowest up to s, subject to τ
% Stage 3: Identical Two-Stage Recovery
$\mathcal{M}_{\text{warm}} \leftarrow \text{SFT}(\mathcal{M}_{\text{pruned}}, \mathcal{D}_{\text{sft}})$ ▷ restore EOS & format
$\mathcal{M}' \leftarrow \text{GRPO}(\mathcal{M}_{\text{warm}}, \mathcal{D}_{\text{rlvr}})$ ▷ recover reasoning
return $\mathcal{M}'$

C Inference Efficiency

Table 5: Inference characteristics at 20% sparsity (clean v2 benchmark, single-GPU, no contention). Pruning reduces parameters by 19% and memory by ∼28%, with TTFT improving 18%. Per-token throughput decreases 19% (0.81×) due to non-uniform layer widths from full-layer removal that current GPU kernels do not optimize.

Configuration	TPS (tok/s)	TTFT (ms)	Per-token (ms)	Params (B)
Dense 7B	72.0	20.1	13.9	7.62
CKA-only 20%	58.3	16.4	16.7	6.16
Standard 20%	58.2	16.4	16.7	6.16

All pruned models achieve ∼28% memory reduction from structural parameter removal. TTFT (time to first token) improves by 18% (20.1→16.4 ms), reflecting faster prefill with fewer parameters. Per-token throughput decreases by 19% (72.0→58.3 tok/s, or 13.9→16.7 ms per token), consistent with non-uniform layer widths creating computational bottlenecks that current GPU kernels do not optimize away. CKA-only and Standard show identical throughput (58.3 vs 58.2 tok/s), confirming that the pruning criterion affects recovery capacity but not inference speed at matched sparsity.

D CKA Coefficient α Ablation

At α=4.0, CKA overlap dominates the composite score, and the resulting pruned model exhibits active GRPO learning: reward progresses from 0.146 to a peak of 0.256 before settling at 0.198, with preliminary GSM8K accuracy of 62.93% (N=774/1,319; SFT baseline 60.71%, Δ=+2.22 pp). This contrasts sharply with α=1.0 (RASP), which collapses to peak reward 0.061, and aligns with the main finding that CKA-guided structure selection enables RL recovery. The α=4.0 result also represents the only condition where $|\Delta| > 1$ pp among fine-grained pruning methods; full-sample evaluation is pending.

Table 6: Effect of CKA coefficient α on post-prune accuracy at 20% sparsity (N=100). α=0.0 reduces to Standard importance-only pruning. Best in **bold**, second-best underlined.

α	Post-Prune Acc. (%)
0.00	27
0.25	23
0.50	22
1.00	31
2.00	<u>36</u>
4.00	**40**

E Extended Training Dynamics

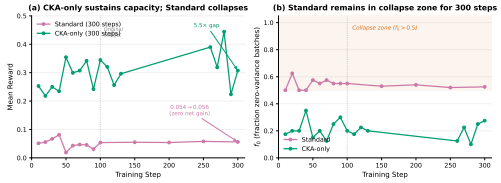


Figure 5: Extended GRPO training (300 steps, 3× original budget). (a) Standard reward shows zero net gain (0.054→0.056); CKA-only oscillates in the 0.22–0.44 range with a 5.5× gap at step 300. (b) Standard f_0 remains in the collapse zone (>0.5); CKA-only stays below 0.30.

F MATH-500 Out-of-Distribution Evaluation

Table 7: Out-of-distribution evaluation on MATH-500 (N=500, greedy decoding, `max_new_tokens=2048`). The recovery capacity spectrum persists across sparsity levels. Reward-accuracy decoupling holds across all completed SFT/GRPO pairs ($|\Delta| \leq 1.2$ pp).

Method	Recovery	MATH-500 (%)
Dense	–	53.0
20% sparsity		
CKA-only	SFT	26.8
CKA-only	GRPO	26.6
Standard	SFT	25.8
RASP	SFT	22.6
25% sparsity		
CKA-only	SFT	19.2
CKA-only	GRPO	18.4
Standard	SFT	14.0
Standard	GRPO	12.8

Over half of Dense model responses are truncated at 2048 tokens, depressing absolute accuracy (73.6% at 4096 tokens vs 53.0% at 2048). All pruned models are evaluated under the same constraint for fair comparison. 20% SFT/GRPO pairs incomplete due to evaluation failures; available checkpoints reported.

G MBPP Code Generation

H CKA Scoring Stability

CKA-based structure rankings are highly stable across calibration data volumes: Spearman ρ between rankings from 256, 512, and 1024 calibration samples ranges from 0.964 to 0.979, indicating that the scoring criterion is not sensitive to calibration set size.

I Seed Reproducibility

CKA-only GSM8K accuracy across 3 GRPO seeds: 64.37% (seed 42), 64.22% (seed 123), 64.58% (seed 456), yielding mean 64.39% ± 0.19 pp. Standard: mean 59.84% ± 0.38 pp across the same 3 seeds. See Table 11 for the full summary.

J Sparsity Cliff at 30%

All tested methods collapse at 30% sparsity: Standard retains 4% accuracy, RASP and CKA-only reach 0%. This capacity threshold, independent of pruning criterion, validates 20% as the operating point and confirms that the recovery capacity spectrum reflects how methods allocate a limited pruning budget.

K Multi-Seed GRPO Results

At 20% sparsity, the 4.55 pp gap between CKA-only and Standard is robust across seeds, with non-overlapping ranges (CKA-only: 64.22–64.58%, Standard: 59.44–60.20%). Standard shows higher seed variance (0.38 pp vs 0.19 pp), consistent with borderline-collapse dynamics where small perturbations produce larger outcome variation. Note: Standard seed 42 was trained from a separately initialized SFT checkpoint (accuracy 59.21% vs. 59.29% for seeds 123/456; Δ = 0.08 pp). At 15% sparsity, the gap widens to 7.20 pp (72.76% vs 65.56%, $t \approx 16.4$, $p < 0.001$), with non-overlapping ranges (CKA-only: 72.10–73.09%, Standard: 65.13–65.43%).

L Cross-Sparsity Validation

At 15% sparsity (3-seed mean), the CKA-only advantage widens to +7.20 pp (72.76% vs 65.56%), compared to +4.55 pp at 20%. CKA-only at 15% (72.76%) represents the highest mean accuracy across all experimental conditions, retaining 80.1% of original dense model performance

Table 8: MBPP code generation accuracy (N=374, greedy decoding). All fine-grained pruning methods converge near a capability floor (∼75% code capability loss vs Dense), limiting differentiation (∼1 pp range, within stderr). ShortGPT shows a larger GRPO delta (+4.01 pp) but at the 95% CI boundary (stderr ≈1.55 pp).

Method	SFT (%)	GRPO (%)	Δ (pp)
Dense	–	53.48	–
CKA-only 20%	12.83	14.17	+1.34
Standard 20%	12.03	13.10	+1.07
RASP 20%	11.76	13.10	+1.34
ShortGPT 20%†	10.43	14.44	+4.01

†Depth pruning; removes entire layers.

Table 9: CKA scoring stability across calibration sizes. Spearman ρ between structure rankings produced by different calibration set sizes. High correlation ($\rho \geq 0.964$) indicates that CKA-based scoring is robust to calibration data volume.

Calibration Sizes	Spearman ρ (min)	Spearman ρ (mean)
256 / 512 / 1024	0.964	0.979

(90.83%). At 25% sparsity (single seed), Standard's SFT accuracy decline accelerates (−7.20 pp from 15%→20%, −8.42 pp from 20%→25%), while CKA-only's decline decelerates (−8.27 pp then −3.79 pp). Reward-accuracy decoupling is preserved ($|\Delta| \leq 1$ pp for both methods) despite Standard's near-zero peak reward (0.008).

M Cross-Architecture Validation

CKA-only pruning causes near-total initial damage on Qwen2.5-Math-7B (1.0% post-prune accuracy vs Standard's 18.5%), yet SFT recovers CKA-only to 51.50% while Standard remains at 21.00%. This post-prune accuracy reversal provides strong evidence that recovery capacity is shaped by preserved network structure rather than starting accuracy. EOS rates remain below 40% for both methods, indicating that SFT warmup is less effective on this architecture; GRPO recovery may yield further improvements but was not run due to compute constraints.

N Structural Overlap Analysis

CKA-only and Standard pruning produce markedly different removal patterns across the 28 layers of DeepSeek-R1-Distill-Qwen-7B at 20% sparsity. CKA-only concentrates pruning in early layers (L0–

Table 10: Sparsity cliff at 30% structured sparsity (N=100). All tested pruning criteria collapse, with near-zero accuracy and EOS generation rates below 6%.

Method	Acc. (%)	EOS Rate (%)
Standard	4	5.5
RASP	0	3.0
CKA-only	0	–†

†Terminated early after collapse confirmed.

Table 11: Multi-seed GRPO results on GSM8K (N=1,319). CKA-only and Standard each trained with 3 GRPO seeds (42, 123, 456); pruning calibration and SFT are identical across all runs within each sparsity level.

Sparsity	Method	Mean Acc. (%)	Std (pp)	Seeds
20%	CKA-only	64.39	0.19	3
	Standard	59.84	0.38	3
	RASP	60.88	–	1
15%	CKA-only	72.76	0.57	3
	Standard	65.56	0.50	3

9 account for 72.5% of removed MLP groups), preserving nearly all structures in later layers (L16–27). Standard distributes removals more uniformly, pruning substantial numbers of groups in late layers (L16–18, L22, L25–26).

The methods share only partial overlap in which structures they remove. 65 MLP groups are pruned exclusively by CKA-only (retained by Standard), concentrated in L6–9 and L27—structures that gradient importance scores as important but CKA identifies as redundant with neighboring groups. Conversely, 46+ groups are pruned exclusively by Standard (retained by CKA-only), concentrated in L16–18, L22, and L25–26—structures that gradient importance scores as unimportant but CKA identifies as representationally unique.

This pattern is consistent with the CKA heatmap (Figure 4): CKA-only's aggressive early-layer pruning disrupts dense representations (low diagonal CKA, μ=0.62) but preserves diverse late-layer pathways, while Standard's uniform pruning maintains dense fidelity (μ=0.90) at the cost of removing unique computational pathways in later layers.

O KL Divergence During GRPO Training

Table 12: Cross-sparsity validation on DeepSeek-R1-Distill-Qwen-7B. GSM8K accuracy (N=1,319) after SFT warmup and GRPO recovery. 15% and 20% GRPO values are mean over 3 seeds (see Tables 2, 11). ‡Single seed. Mean $|\Delta| \leq 1$ pp across all conditions.

Sparsity	Method	SFT (%)	GRPO (%)	Δ (pp)
15%	CKA-only	72.18	**72.76±0.57**	+0.58
	Standard	66.49	65.56±0.50	−0.93
20%	CKA-only	63.91	64.39	+0.48
	Standard	59.29	59.84	+0.55
25%‡	CKA-only	60.12	59.59	−0.53
	Standard	50.87	51.86	+0.99
30%	All	Collapse (≤4%; Table 10)		

Table 13: Cross-architecture validation on Qwen2.5-Math-7B (Yang et al., 2024) at 20% structured sparsity. Accuracy evaluated on GSM8K (N=200). Post-SFT reports mean ± std over multiple SFT runs. EOS rate: fraction of generations that produce a valid end-of-sequence token.

Method	Post-Prune (%)	Post-SFT (%)	EOS Rate (%)
CKA-only	1.0	**51.50 ± 3.53**	27.0
Standard	18.5	21.00 ± 2.88	36.5

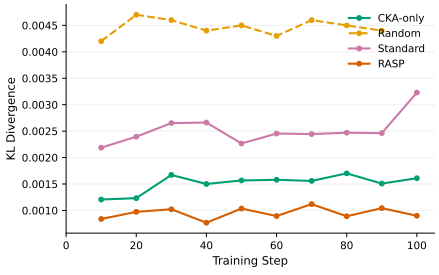


Figure 6: KL divergence between the policy and reference model during GRPO training. RASP maintains near-zero KL (<0.001), consistent with minimal policy movement during collapse. Random shows the highest KL (∼0.004–0.005), indicating more policy exploration. CKA-only and Standard fall between these extremes.